

altairsim — Quick Reference

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Getting out, and back in

Key	Does
<code>^E</code>	ATTN — stop the machine and take the keyboard back. Nothing is lost.
<code>RUN</code>	Resume, at the exact instruction it stopped on.
<code>QUIT</code>	Leave. (There is no <code>EXIT</code> .)

Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values (`SET mem0 fill=` then `Tab`); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.altairsim_history`.

Key	Does
<code>Ctrl-A</code> / <code>Ctrl-E</code> (or <code>Home</code> / <code>End</code>)	start / end of line
<code>Alt-B</code> / <code>Alt-F</code> (or <code>Ctrl-Left</code> / <code>Ctrl-Right</code>)	back / forward one word
<code>Ctrl-W</code> / <code>Ctrl-K</code> / <code>Ctrl-U</code>	erase word behind / to end of line / whole line
<code>Backspace</code> / <code>Delete</code>	erase before / under the cursor

Command line

```
altairsim [options] [machine]
```

machine	a built-in name, or a config file (has a '/' or ends .toml). Omitted: ./altairsim.toml if there is one, else `default`.
-m, --machine <n>	ALWAYS a built-in name -- never a file.
-f, --file <path>	ALWAYS a file -- never a built-in name.
-n, --none	empty backplane: no boards, no memory, nothing.
-l, --list	list the built-in machines and exit.
-s, --script <f>	run a command script, then exit with its status.
-x, --exec <cmd>	run one monitor command (repeatable), then exit.
-i, --interactive	after --script/--exec, stay in the monitor.
--mcp	MCP server on stdio.
-v, --version	print the version and exit.
-h, --help	print this help and exit.

Monitor commands

Type the part before the bracket.

Command	Does	Usage
BO[ARDS]	List, add, or remove boards on the backplane.	BOARDS [LIST] ADD <type> <id> [k=v...] REMOVE <id>
B[REAK]	Set a breakpoint on an address, memory/I/O access, or tape stop.	BREAK [<addr> MEM R W <addr> IO R W <port> TAPE STOP] [IF <expr> LOADS <expr>] [TRACE ON OFF]
COM[PARE]	Compare a range of memory against another address.	COMPARE <range> <addr>
C[ONFIG]	Load or save the whole machine as a TOML file.	CONFIG LOAD <f.toml> CONFIG SAVE <f.toml>
CONN[ECT]	Attach a serial unit to an endpoint (console, socket, file, ...).	CONNECT <id>:<u> <endpoint>
CONS[OLE]	Show or set the host console's properties.	CONSOLE [<k>=<v>...]
DE[POSIT]	Write bytes into memory at an address.	DEPOSIT <addr> <bytes...>
DI[SASM]	Disassemble memory into instructions.	DISASM [<addr> <range>] [n] [CPU=8080]
DISC[ONNECT]	Unplug the endpoint from a serial unit.	DISCONNECT <id>:<u>
D[UMP]	Show memory as hex and ASCII.	DUMP [<addr> <range>] [WIDTH=16]
E[EDIT]	Enter bytes into memory interactively from an address.	EDIT <addr> [ROM]
EX[AMINE]	Point the front panel at an address (and show that byte).	EXAMINE [<addr>]
F[ILL]	Fill a range of memory with a byte.	FILL <range> <byte>
HE[LP]	Show help for a command.	HELP [<command>]
H[ISTORY]	Replay the recent instruction (or bus-cycle) history.	HISTORY [BUS CPU] [n]
I[N]	Read a byte from an I/O port.	IN <port>
L[OAD]	Load a file into memory (binary or Intel hex).	LOAD <file> [AT <addr>] [FORMAT=BIN HEX] [ROM]
M[OUNT]	Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it.	MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...]
MOV[E]	Copy a range of memory to another address.	MOVE <range> <dest> [ROM]

N[EXT]	Step one instruction, running any CALL/RST to completion.	NEXT
NO[BREAK]	Remove a breakpoint, or all of them.	NOBREAK [id]
O[UT]	Write a byte to an I/O port.	OUT <port> <byte>
P[OWER]	Power-cycle the machine -- the only thing that clears RAM.	POWER
Q[UIT]	Leave the simulator.	QUIT
REGI[ON]	Add a RAM or ROM region to a memory board.	REGION ADD <id> type=ram rom at=<addr> [size= mount=]
RE[GS]	Show the CPU registers (SET REG changes one).	REGS SET REG <r>=<v>
RES[ET]	Reset the machine, keeping RAM (RESET CPU resets just the processor).	RESET [CPU]
REST[ORE]	Load machine state back from a snapshot.	RESTORE <file>
R[UN]	Start or resume the machine, optionally at an address.	RUN [addr]
SA[VE]	Write a range of memory out to a file.	SAVE <file> <range> [FORMAT=BIN HEX OCTAL PRN]

SEARCH	Find bytes or a string in a range of memory.	SEARCH <range> <bytes...> "str"
SET	Change a property of a board, the console, display, a register, or the bus.	SET <id>[:<u>] CONSOLE DISPLAY REG BUS <k>=<v>
SHOW	Display the state of a board, the bus, or the machine.	SHOW <id> BOARDS BOARD <type> [UNITS] MACHINES MACHINE [<name>] BUS [MAP IO IRQ CONTENTION] ROMS MOUNTS PATHS CONSOLE DISPLAY SYMBOLS VERSION
SNAPSHOT	Save the whole machine state to a file.	SNAPSHOT <file>
STEP	Run one instruction (or n), showing the registers after each.	STEP [n]
SYMBOLS	Load or clear a symbol table for disassembly.	SYMBOLS LOAD <file> [REPLACE] SYMBOLS CLEAR
TRACE	Log every bus cycle while the machine runs.	TRACE ON OFF [file] [MASK=IN,OUT,IRQ,DMA,CONTENTION]
TYPE	Feed text to the guest as if typed at its keyboard.	TYPE "text"
UNMOUNT	Take a disk or tape out of a drive.	UNMOUNT <id>:<u>
WHO	Say which board answers an address or I/O port.	WHO <addr> WHO IO <port>

Boards

CPU

Type	What it is
6800	Altair 680b CPU board: a Motorola 6800 at 500 KHz. Decodes nothing -- it drives the bus. The 88-CPU's twin, one core down, with memory-mapped I/O
8080	MITS 88-CPU: an 8080A at 2 MHz. Decodes nothing -- it drives the bus
8085	Generic 8085 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with an 8085 core (RIM/SIM + TRAP/RST 5.5/6.5/7.5)
z80	Generic Z80 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with a Z80 core

Memory

Type	What it is
bankmem	S-100 bank-switched RAM. One card, four decoders (card=vector cromemco64kz northstar expandoram2): a write-only select port swaps which RAM plane(s) drive the bus. Each card owns its own decode -- one-hot select (Vector 40), 8-bit bank mask (Cromemco 40), on/off+one-hot toggle (North Star C0), or PROM page-select (ExpandoRAM II FF, approximated)
memory	RAM/ROM board: a list of regions and PHANTOM* -- plain, unbanked memory (bank switching is its own board, bankmem)
v2z80rom	S100Computers V2 Z80 CPU board -- its onboard paged monitor EEPROM (the Z80 itself is board 'z80cpu'). An 8K 28C64 at F000-FFFF holding two 4K pages, builtin:master0 (low) / master1 (high), selected by OUT D3H bit1 (bit0=1 inactivates the EEPROM so RAM shows through). Shadows RAM in its window while enabled. Cold-start the MASTER monitor with startup=["RUN F000"]; the 'I' command boots CP/M 3 off a dualsd card

Disk

Type	What it is
16fdc	Cromemco 16FDC: WD FD1793 soft-sector floppy (single + double density), up to 4 drives. Disk registers at 30-34, a TMS 5501 console UART at 00-09 (unit 'tty'), and a 4K RDOS 2.52 boot PROM at C000 (OUT 40H banks it out, RESET restores it). Boots CDOS
64fdc	Cromemco 64FDC: the 16FDC's 1983 successor -- same FD1793 + TMS 5501, carrying an 8K RDOS 3.12 boot PROM at C000-DFFF (OUT 40H banks it out, RESET restores it). Boots CDOS
dcdd	MIT 88-DCDD: 8" hard-sector floppy, up to 16 drives. Three ports at BASE+0..2. INVERTED status bits
dualide	S100Computers IDE-AB (CF): the IDE/CompactFlash half of the IDE+ESP32 combination board -- two CF sockets (drives 0/1 = A:/B:) for CP/M 3. Five 8255 ports at BASE+0..4 (default 30): A/B data, C control lines, mode config, drive select. Programmed-I/O ATA register engine (LBA read/write, 512-byte sectors). No boot PROM -- the CPU board's monitor boots CP/M from the CF. Mounts the SAME card image as dualsd (a .img with a .geo geometry sidecar); pair with dualsd for the full A:/B:/C:/D: system
dualsd	S100Computers Dual SD: two microSD sockets (drives 0/1) presented as raw 512-byte-sector CF/SD cards, for CP/M 3. Two ports at BASE+0..1 (default 80): status/command + data. Programmed-I/O command/handshake engine (33H-lead + 8 commands). No boot PROM -- the CPU board's monitor loads CP/M from track 0. Mount a card image (a .img with a .geo geometry sidecar)
hdsk	MIT 88-HDSK Datakeeper: Pertec hard disk, 256-byte sectors from a linear .DSK. Eight ports at BASE+0..7 (default A0). Command/handshake protocol, four page buffers
icom	iCOM FD3712/FD3812 8" floppy: a programmed-I/O command/handshake controller on the S-100 Interface board. Two ports at BASE+0..1 (default C0) plus a boot PROM and 6810 scratch RAM in high memory (rom=builtin:icom-fd3712-cpm icom-fd3712-fdos icom-fd3812-cpm). Boots CP/M 2.2 (single and double density) and FDOS. Up to 4 drives
mds	MIT 88-MDS: 5.25" minidisk, 4 drives. Same three ports as the dcdd -- but 300 RPM, 64 us/byte, and a motor that stops after 6.4 s
tarbell	Tarbell #1011: single-density WD FD1771 floppy, up to 4 drives. Eight ports at BASE+0..7 (default F8). Carries a 32-byte boot PROM that shadows 0000 over PHANTOM* -- boots CP/M automatically at reset (bootstrap=on)
tarbelldd	Tarbell #2022: double-density WD FD1791 floppy (mixed-density media, SD track 0), up to 4 drives. The single-density card's twin with a bitmap OUT-FC latch and a port-FD DMA/ext-addr register. Same 32-byte boot PROM
versafloppy	SD Systems VersaFloppy I/II: WD FD177x soft-sector floppy, up to 4 drives. Eight ports at BASE+0..7 (default 60). variant=vfi (FD1771, single density) vfii (FD1791, single+double). Boots SDOS with the SBC-200 + DDBIOS

Serial

Type	What it is
2sio	MITs 88-2SIO: two 6850 ACIAs, units 'a' and 'b'. Four ports at BASE+0..3
680io	Altair 680b onboard I/O: a 6850 ACIA console ('tty') at F000/F001 and the config-strap read port at F002. Memory-mapped
pmmi	PMMI MM-103: Bell 103 modem on an S-100 card, unit 'line'. Four ports at BASE+0..3 (default C0), read/write different registers. Transmit/receive over a ByteStream; CONNECT it to in:/out: files. No dialer; modem status is a fixed stub
propio	S100Computers Console IO Board (Parallax-Propeller console), unit 'serial'. A usio subtype preset to the board's documented convention: status/data at 00/01, RX-ready = status bit 1, TX-ready = status bit 2, both active high. Every strap (status_port/data_port/rdr_bit/tdre_bit/polarity) is still overridable -- the real board is jumpered. Polled, no interrupts. CONNECT it to a file, socket, serial port, in:/out:
sbc	SD Systems SBC-100/200: Z80 single-board computer. One 8-port block (78-7F): Intel 8251 console (unit 'tty', data 7C / status 7D, RxD->/DSR auto-baud for MSMONR21), Z80-CTC (78-7B) whose ch1 raises a mode-2 keyboard interrupt (vector 0x82) off the 8251 RxRDY, and a parallel port (7E/7F) whose OUT 7F bit 1 switches the onboard PROM out. Optional onboard boot PROM via [[board.socket]] (at+mount). variant=sbc100 sbc200
sio	MITs 88-SIO: one COM2502 UART, unit 'tty'. Two ports at BASE+0..1. INVERTED status bits
turnkey	MITs 8800b Turnkey Module: phantom boot PROM (FC00-FFFF), integrated 6850 SIO (unit 'tty', default 0x10), sense switches at FF, and the Auto-Start JMP jam. Sockets via [[board.socket]]
usio	Universal Serial board: a UART-agnostic serial card, unit 'serial'. Two ports you strap: a status/control port (status_port -- read synthesizes RDR/TDRE at bit positions you pick, write is discarded) and a data port (data_port). Built-in profiles preset the straps: profile=tuart (Cromemco TU-ART) imesai-sio2 compupro-if2 (CompuPro Interfacer II) compupro-ss1 (CompuPro System Support 1). Polled, no interrupts. CONNECT it to a file, socket, serial port, in:/out:

Tape

Type	What it is
680kcacr	Altair 680b KCACR audio-cassette interface: a 1602-family UART recording Kansas City Standard FSK, memory-mapped at F010 (status/control) and F011 (data), active-LOW. Adds software motor control (control D7=on, D6=off) and interrupt-driven transfer (D0/D1 enables pull the 6800 IRQ). Reuses the 88-ACR tape machinery -- MOUNT a tape, WIND/REWIND it
acr	MITs 88-ACR: cassette. An 88-SIO B + an FSK modem, unit 'tape'. Brings the WIND/REWIND/EXTRACT verbs and a tape counter
uio	MITs 88-UIO: serial + cassette on one board. A 6850 (unit 'serial', default 0x10) and an 88-ACR cassette section (unit 'tape', default 0x06) with motor control and a SW-1 MITs/Kansas-City modulation switch. Defaults reproduce the standard 0x10 + 0x06 layout

Parallel and printer

Type	What it is
4pio	MIT 88-4PIO: up to four 6820 PIAs, sections ja/jb.. per port. 16 ports from BASE (default 20). Software-set direction; CONNECT each section
680uio	Altair 680b Universal I/O: a second 6850 ACIA serial port ('serial') and a 6820 PIA parallel port (sections 'p1a/p1b', 'p2a/p2b' with pias=2) in an S9-relocatable window (default base F000: serial F006/F007, PIA F008-F00F), plus fixed switch inputs at F003 and a non-latched output at F010-F013. Memory-mapped, active-high
c700	MIT 88-C700: Centronics line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Output-only; CONNECT it to a file, a socket, or a real printer queue
d7a	Cromemco D+7A: analog + parallel I/O. Eight ports from BASE (default 18): one parallel port + seven two's-complement A/D-in/D/A-out channels. Reads 1-2 JS-1 joysticks from the host
lpc	MIT 88-LPC: 88-LP line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Line-buffered: 6-bit codes + PRINT/LINE FEED/CLEAR. CONNECT it to a file, a socket, or a real printer queue
pio	MIT 88-PIO: 8-bit parallel port, units 'out'/'in'. Two ports at BASE+0..1 (default 04). CONNECT a printer, a keyboard, a socket

Video

Type	What it is
dazzler	Cromemco Dazzler: color graphics from a framebuffer in main RAM. Two ports at BASE+0..1 (default 0E): control/status and format. 32x32 to 128x128, 16 colors/greys. Needs a Display
vdb8024	SD Systems VDB-8024: an 80x24 video terminal on one board -- the video console for an SBC-100/200 (the alternative to the 8251). Two I/O ports at BASE+0..1 (default 00): status/keyboard/display. Unit 'keyboard' (CONNECT). Optional keyboard-strobe interrupt strap (interrupt=vi0..vi7) for the SBC-200's CTC to vector - what the SD video CBIOS needs; polled by default. Boots sdmonv21. Needs a Display
vdm1	Processor Technology VDM-1: memory-mapped 16x64 video, screen RAM at BASE (default CC00), scroll/status port (default CC). Needs a Display

Systems

Type	What it is
sol	Processor Technology Sol-PC I/O: serial, keyboard, parallel, CUTS tape as one board. Seven ports F8..FE. Units serial/printer/keyboard (CONNECT) and tape1/tape2 (MOUNT). Brings the WIND/REWIND/EXTRACT verbs and a tape counter

Other

Type	What it is
fp	Altair front panel: the SENSE switches a guest reads at IN 0FFH -- a configured byte (SET fp0 sense= or TOML), not toggled here. No OUT
hostbridge	Host Bridge: guest <-> host file transfer, sandboxed. OUR OWN BOARD, not a period one. Two ports at BASE+0..1. R.COM/W.COM/HDIR.COM
ss1	CompuPro System Support 1: multifunction S-100 board. Today the OKI MSM5832 battery-backed real-time clock/calendar (command/data at base+10/+11; base default 50H). The 2651 UART, 8253 timer and dual 8259A interrupt controllers are being added in phases; the 9511/9512 math socket is unpopulated
virtc	MIT 88-VI/RTC: vectored interrupts (VI0-VI7 -> RST n) and a real-time clock. One port at FE

Machines

Machine	What it is
8085	A minimal 8085 machine: an 8085 CPU, 64K of RAM, and a 2SIO console.
acuter	ACUTER at F000 -- CUTER on a plain Altair, with a terminal instead of a VDM.
altair680	The Altair 680b -- MITS's second machine, and a different animal from the 8800.
altmon	An Altair with a monitor in ROM and a terminal on it.
amon	AMON 3.1 in a 4K EPROM at F000 -- Martin Eberhard's full-featured Altair monitor.
bankmem	A bank-switched RAM machine: a Z80, a console, and a Vector Graphic 64K bankmem.
basic4k	The machine Altair 4K BASIC was sold to run on: an 88-SIO Teletype, a cassette in the ACR.
basic8k	The machine Altair 8K BASIC was sold to run on: an 88-2SIO terminal, a cassette in the ACR.
cdbl	The default machine with the Combo Disk Boot Loader in the PROM socket.
compupro	A stock Altair with a CompuPro System Support 1 board for its clock/calendar.
cuter	CUTER 1.3 driving a Processor Technology VDM-1 -- the real Sol/CUTS monitor.
dazzler	A Cromemco Dazzler in an Altair -- the bench for the S-100's first color graphics card.
default	The machine you get when you name none: 56K, and the DBL boot PROM at FF00.
dualide	S100Computers "IDE-AB CF" machine -- a V2 Z80 CPU board booting CP/M 3 off a CompactFlash card.
dualidesd	S100Computers "IDE-AB CF+ESP32" machine -- both boards, CP/M 3 on CF drives A:/B: and SD drives C:/D:.
dualsd	S100Computers "Dual SD" machine -- a V2 Z80 CPU board booting CP/M 3 off a microSD card.
icom	iCOM FD3712 8" floppy machine -- boots CP/M 2.2 off a single-density iCOM disk.
lineprinter-lpc	The default machine with an 88-LPC line printer at port 02; CONNECT lpt0:prn to a file or the console.
lineprinter	The default machine with an 88-C700 line printer at port 02; CONNECT lpt0:prn to a file or the console.
minidisk	The Altair Minidisk: an 88-MDS at 08 and the MDBL boot PROM. You supply the 5.25" disk.
original	The Altair as it actually left Albuquerque.
parallel	The default machine with two MITS parallel boards: an 88-PIO and an 88-4PIO.
ps2	The machine MITS Programming System II ran on: basic8k's cards, but not basic8k's bootstrap.
ps2int	MITS Programming System II, WITH INTERRUPTS. ps2 with A9 down and an 88-VI/RTC in it.
rombasic	MITS Extended ROM BASIC 16K -- Extended BASIC that runs directly out of ROM.

sbc200	SD Systems SBC-200 -- a 4 MHz Z80 single-board computer running the MSMONR21 monitor.
sbc200v	The SD Systems SBC-200 with a VDB-8024 VIDEO console: SDMONV21 on an 80x24 screen.
sol20	The Processor Technology Sol-20 -- an integrated 8080 machine, running SOLOS.
tarbell	Tarbell #1011 single-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.
tarbelldd	Tarbell #2022 double-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.
turnkey	The MITS 8800bt -- an Altair with a Turnkey Module where the front panel used to be.
vdm1	A Processor Technology VDM-1 in an Altair, and a demo that draws on it.
z80	A minimal Z80 machine: a z80 CPU, 64K of RAM, and a 2SIO console.

A machine file, in one look

```
[machine]
name      = "mine"
base      = "default"      # start from a machine, and say what is DIFFERENT
startup   = ["RUN FF00"]    # the operator's own keystrokes. There is no BOOT verb.

[[board]]                # type + a NEW id      -> ADD the card
type       = "2sio"        # type + an id from the base -> REPLACE it outright
id         = "sio0"        # NO type + an id      -> MODIFY the one already there
port       = 10            # remove = true       -> PULL THE CARD OUT

[board.unit.a]            # a unit's own settings
connect    = "console"

[[board.region]]          # a list the card owns (memory)
type       = "ram"
at         = 0000          # hex: it is an address
size       = "56K"         # decimal: it is a size

[[board.drive]]           # a list the card owns (disk controllers)
unit       = 0
mount      = "cpm.dsk"     # relative to THIS FILE

[console]                 # the HOST's terminal -- not a board
strip7out  = true
base       = octal         # read/print the wire class in split octal (MITS style)
```

Paths: a path *inside* a machine file is relative to **that file**. A path you type is relative to **your shell**.

Endpoints — `CONNECT <id>:<unit> <endpoint>`

Endpoint	Is
<code>console</code>	the host terminal. Exactly one unit may hold it.
<code>null</code>	nowhere. Writes vanish, reads never come.
<code>loopback</code>	itself — what you write comes back.
<code>scripted</code>	a caller in place of a human — what MCP and the tests type into.
<code>socket:PORT</code>	listens — this is telnet-in.
<code>socket:HOST:PORT</code>	calls out .
<code>serial:DEVICE</code>	a real serial port on this host.
<code>in:PATH</code>	a host file as a reader (paper tape). <code>?cps=N</code> paces it.
<code>out:PATH</code>	a host file as a punch — 8-bit clean, never truncating.
<code>terminal</code>	a window the simulator draws itself (SDL builds). <code>?emulation=vt100 adm3a vt52 h19, ?size=COLSxROWS</code> .
<code>printer:QUEUE</code>	a real print queue on this host.
<code><endpoint> FILE</code>	a tap: append <code> FILE</code> to any endpoint to also log the line.